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Dear Editor,

I am pleased to submit our original research article, “Graph Structure Learning for Traffic Prediction”, for consideration for publication in the *International Journal of Data Science and Analytics*. Traditional graph-based methods for traffic forecasting rely on predefined adjacency matrices based on physical road proximity, which often fail to capture true functional dependencies between roads. In this work, we introduce a knowledge discovery framework that learns the underlying graph structure directly from traffic data using continuous optimization, shifting from heuristic graph construction to data-driven knowledge extraction. Key contributions are as follows:

- A novel approach that infers meaningful, persistent traffic dependencies directly from data, overcoming the limitations of predefined spatial relationships.
- Two methodological variants: GSL (directed acyclic graph capturing causal temporal influences) and cGSL (symmetrized cyclic version for spatial models).
- Comprehensive validation on two real-world datasets (SZ-Taxi, Los-loop) demonstrating significant improvements—up to 24.7% RMSE reduction for GCN-cGSL and 21.8% for TGCN-GSL.

This work aligns well with the journal’s scope, advancing both traffic prediction and graph-based knowledge discovery. It shares common ground with recent contributions such as Chen et al. (2026) on trend-aware spatio-temporal graph convolutional networks. While a preliminary version was presented at a national conference (Amintoosi, 2024), this submission represents a substantial extension:

- **Scope:** Extends both GCN and T-GCN architectures (conference focused only on T-GCN).
- **Implementation:** Complete modern implementation in PyTorch 2.0 with Google Colab notebooks for reproducibility available at <https://github.com/mamintoosi/TGCN-GSL-PyTorch> (conference used deprecated TensorFlow 1.x).
- **Experiments:** Comprehensive evaluation on two datasets across multiple horizons (conference presented only preliminary results).
- **Novelty:** Introduces the cyclic cGSL variant and provides detailed ablation studies absent from the conference version.

Sincerely,

Mahmood Amintoosi

References

- Amintoosi, M., 2024. Urban traffic prediction with learning based graph convolutional networks, in: Proceedings of the 55th Annual Iranian Mathematics Conference, Ferdowsi University of Mashhad. pp. 145–148. URL: <https://github.com/mamintoosi/TGCN-GSL-TF>. (In Persian).
- Chen, G., Chen, L., Li, Z., Wei, Y., He, Z., Yang, F., 2026. Ta-stgcn: trend-aware attention spatio-temporal graph convolutional network for traffic flow prediction. *International Journal of Data Science and Analytics* 22, 56. doi:[10.1007/s41060-026-01040-w](https://doi.org/10.1007/s41060-026-01040-w).
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